

# Stability

## Chapter 6

# Requirements on the design of control systems

- Transient Response
- Stability
- Steady state error

# Stability an important performance requirement

There are frequently asked questions about stability that you need to answer. Some of the questions might be:

- Is this closed loop system stable?
- For what range of gains is this system stable?
- How stable is this closed-loop system?

# Stability in terms of total response

- Bounded input bounded output (BIBO) stability
- A system is stable if *every* bounded input yields a bounded output.  $|x(t)| \leq M_x < \infty$
- A system is unstable if *any* bounded input yields an unbounded output.

# Stability in terms of natural response

- for linear, time-invariant systems total response is sum of forced and natural response

$$c(t) = c_{forced}(t) + c_{natural}(t)$$

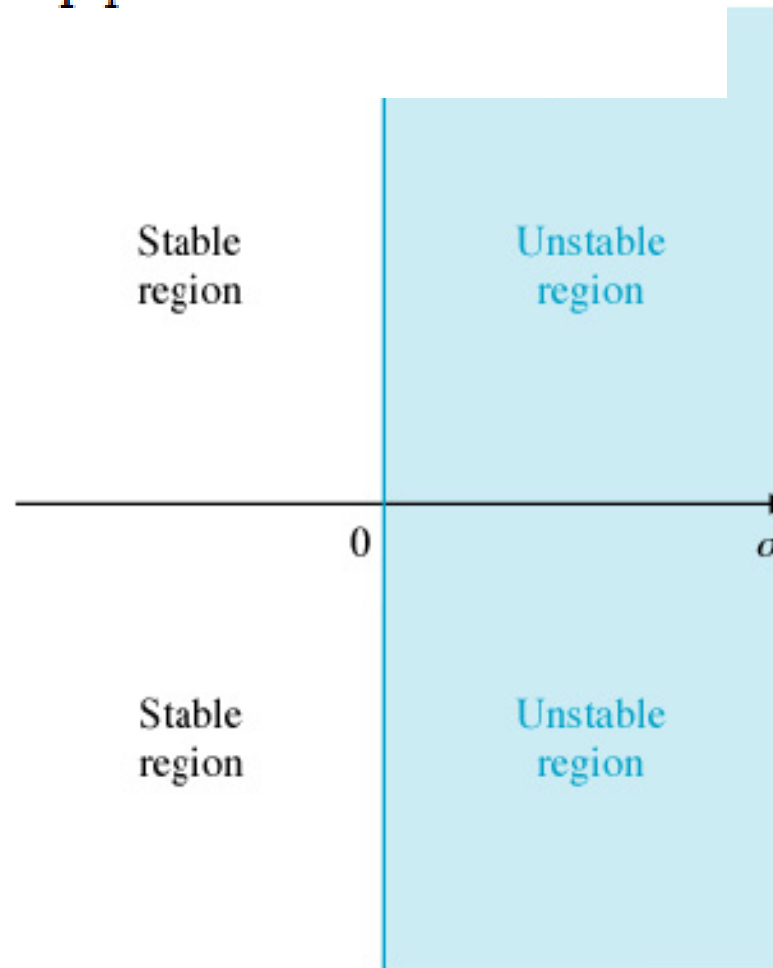
- A system is ***stable*** if the natural response approaches zero as time approaches infinity.
- A system is ***unstable*** if the natural response grows without bound as time approaches infinity.
- A system is ***marginally stable*** if the natural response neither decays nor grows but remain constant or oscillates as time approaches infinity.

# Stability using closed loop transfer function

- Stable systems have closed-loop transfer functions with poles only in the *left half-plane*
- Unstable systems have closed-loop transfer functions with at least one pole in the right half-plane and/or poles of multiplicity greater than 1 on the imaginary axis.
- Marginally stable systems have closed-loop transfer functions with only imaginary axis poles of multiplicity 1 and poles in the left half-plane.

# Stable and unstable regions in s-plane

- A closed-loop control system is stable if and only if all closed-loop poles lie in the left-half  $s$ -plane.



# Sufficient Condition for instability

- If
  - All signs of the coefficients of the denominator of the closed-loop transfer function are not the same
  - powers of  $s$  are missing (any coefficient is zero)then the system is unstable or at best, marginally stable

Closed-loop transfer function denominator polynomials

$a) s^4 + 4s^3 + 20s + 15 = 0$                       unstable

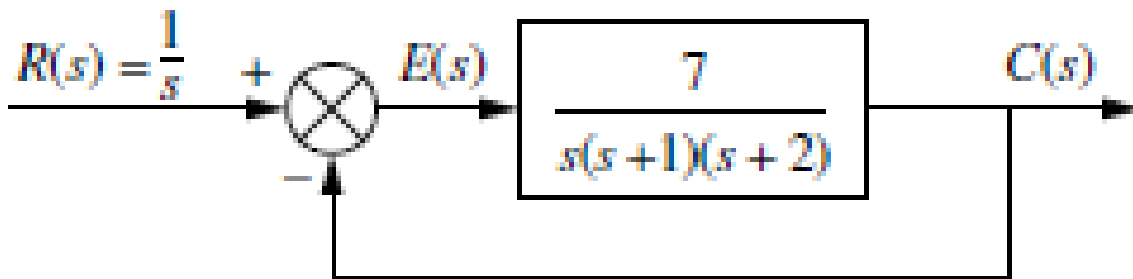
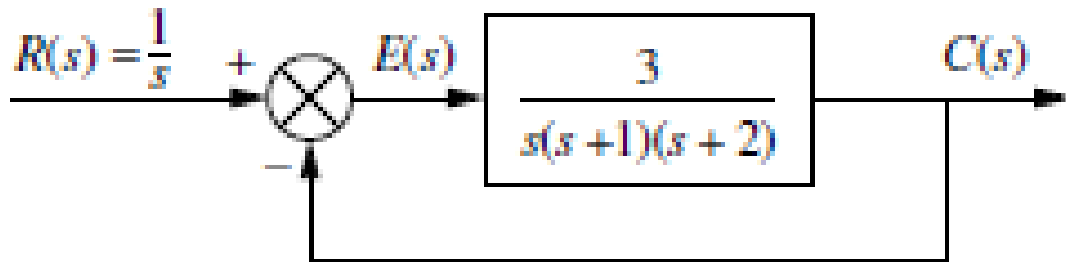
$b) s^4 + 4s^3 + 8s^2 - 20s + 15 = 0$                       unstable

$c) s^4 + 4s^3 + 8s^2 + 20s = 0$                       unstable

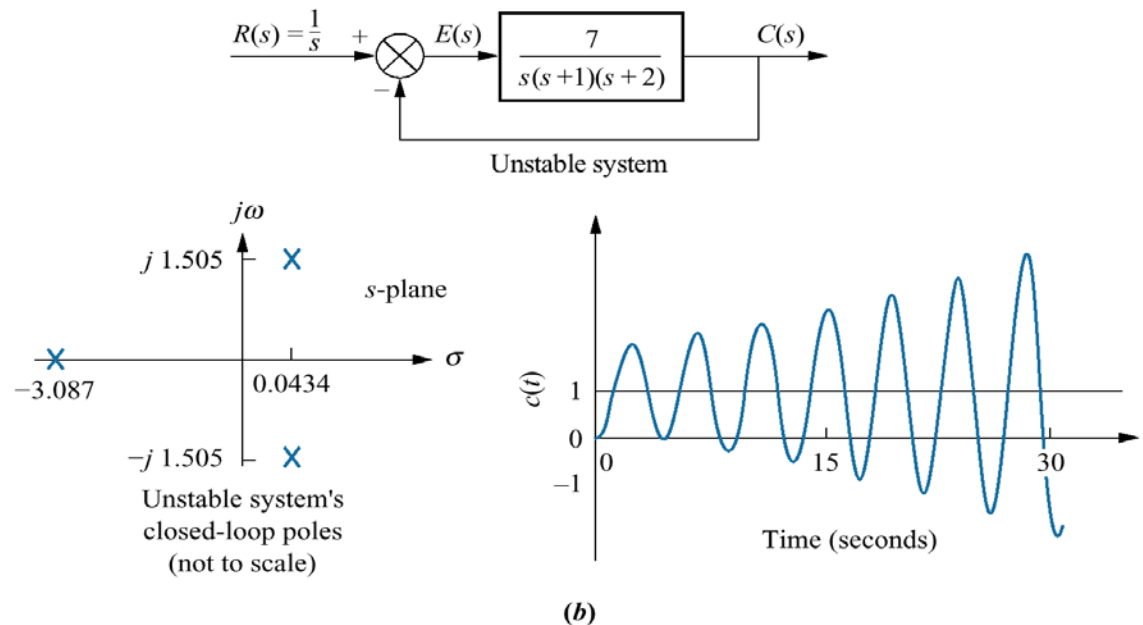
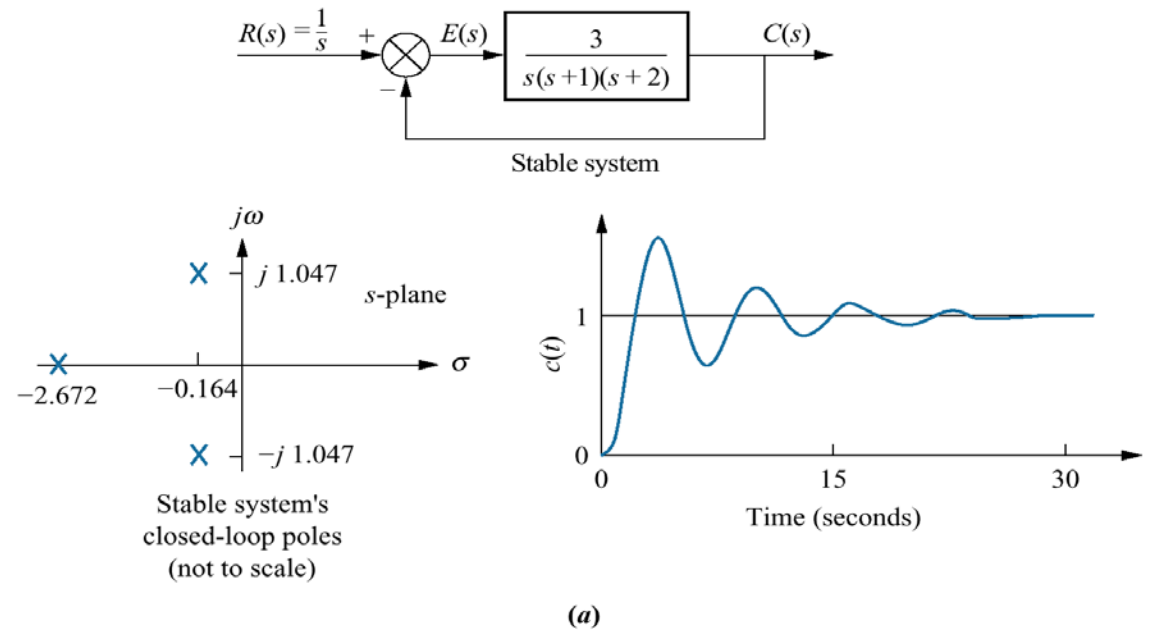
$d) s^4 + 4s^3 + 8s^2 + 20s + 15 = 0$                       uncertain, check roots



Find the location of closed loop poles for the following systems and identify which of these is stable



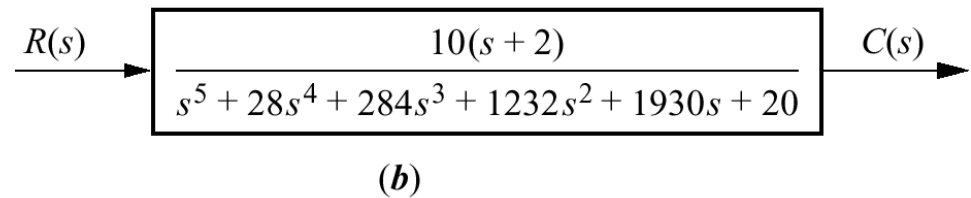
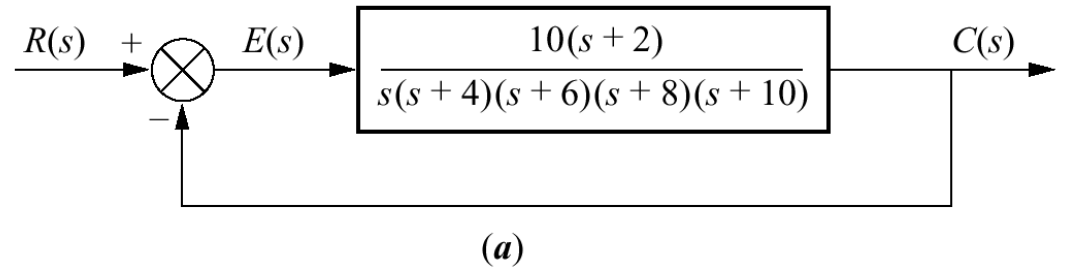
**Figure 6.1**  
 Closed-loop poles  
 and response:  
**a.** stable system;  
**b.** unstable system



## Figure 6.2

Common cause  
of problems in  
finding closed-loop  
poles:

- a.** original system;
- b.** equivalent  
system



# Routh Hurwitz Stability Criterion

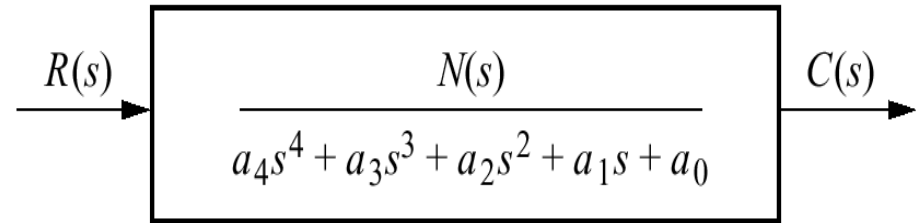
- **Routh stability criterion** is a technique available that can give us information on absolute stability and range of gains for stability.
- Routh criterion tells us how many unstable roots (in the RHP) there are in a polynomial with constant real coefficients, without actually solving for the roots.
- Routh criterion is best suited for equations with at least one unknown coefficients.

## **Procedure:**

- 1) Write the characteristic equation
- 2) If any of the coefficients are zero or negative in the presence of at least one positive coefficient, there is a root or roots that are in the RHP.
- 3) If all coefficients are positive (the necessary condition of stability, but not sufficient) arrange the polynomial coefficients in an array of the following form (the Routh array)

## Figure 6.3

Equivalent closed-loop transfer function



**Table 6.1:** Initial layout for Routh table

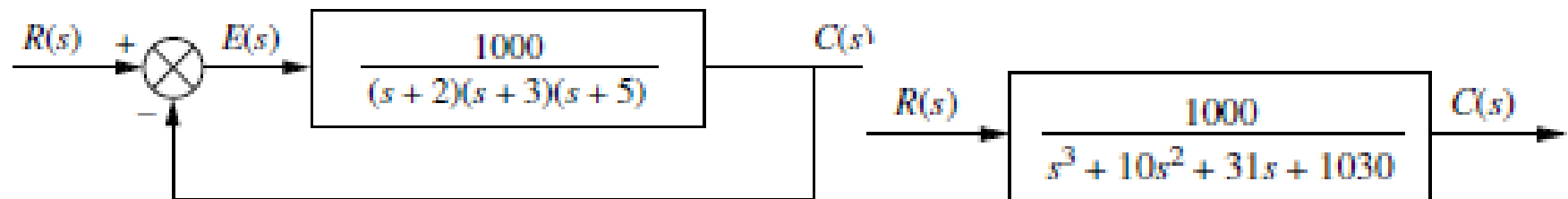
|       |       |       |       |
|-------|-------|-------|-------|
| $s^4$ | $a_4$ | $a_2$ | $a_0$ |
| $s^3$ | $a_3$ | $a_1$ | 0     |
| $s^2$ |       |       |       |
| $s^1$ |       |       |       |
| $s^0$ |       |       |       |

**Table 6.2**  
Completed Routh table

|       |                                                                           |                                                                         |                                                                     |
|-------|---------------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------|
| $s^4$ | $a_4$                                                                     | $a_2$                                                                   | $a_0$                                                               |
| $s^3$ | $a_3$                                                                     | $a_1$                                                                   | 0                                                                   |
| $s^2$ | $-\frac{\begin{vmatrix} a_4 & a_2 \\ a_3 & a_1 \end{vmatrix}}{a_3} = b_1$ | $-\frac{\begin{vmatrix} a_4 & a_0 \\ a_3 & 0 \end{vmatrix}}{a_3} = b_2$ | $-\frac{\begin{vmatrix} a_4 & 0 \\ a_3 & 0 \end{vmatrix}}{a_3} = 0$ |
| $s^1$ | $-\frac{\begin{vmatrix} a_3 & a_1 \\ b_1 & b_2 \end{vmatrix}}{b_1} = c_1$ | $-\frac{\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$     | $-\frac{\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$ |
| $s^0$ | $-\frac{\begin{vmatrix} b_1 & b_2 \\ c_1 & 0 \end{vmatrix}}{c_1} = d_1$   | $-\frac{\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$     | $-\frac{\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$ |

# Creating a Routh table

**PROBLEM:** Make the Routh table for the system shown in Figure



**TABLE 6.3** Completed Routh table for Example 6.1

|       |                                                                       |                                                                   |                                                                   |
|-------|-----------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------|
| $s^3$ | 1                                                                     | 31                                                                | 0                                                                 |
| $s^2$ | 1                                                                     | 103                                                               | 0                                                                 |
| $s^1$ | $-\frac{\begin{vmatrix} 1 & 31 \\ 1 & 103 \end{vmatrix}}{1} = -72$    | $-\frac{\begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix}}{1} = 0$     | $-\frac{\begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix}}{1} = 0$     |
| $s^0$ | $-\frac{\begin{vmatrix} 1 & 103 \\ -72 & 0 \end{vmatrix}}{-72} = 103$ | $-\frac{\begin{vmatrix} 1 & 0 \\ -72 & 0 \end{vmatrix}}{-72} = 0$ | $-\frac{\begin{vmatrix} 1 & 0 \\ -72 & 0 \end{vmatrix}}{-72} = 0$ |

# Interpreting the basic Routh table

- Number of roots in the right half  $s$ -plane is equal to the number of sign changes.
- No sign change means stable system.



Example 6.1:

**TABLE 6.3** Completed Routh table for Example 6.1

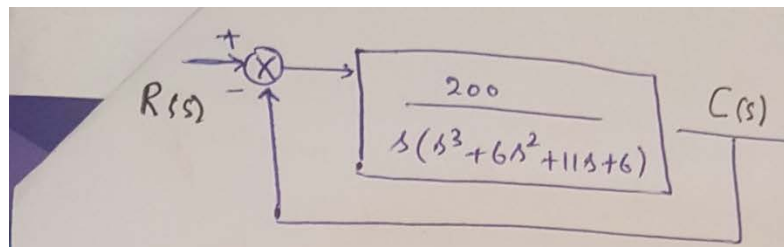
|       |                                                                       |                                                                   |                                                                   |
|-------|-----------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------|
| $s^3$ | 1                                                                     | 31                                                                | 0                                                                 |
| $s^2$ | <del>10</del> 1                                                       | <del>1030</del> 103                                               | 0                                                                 |
| $s^1$ | $\frac{-\begin{vmatrix} 1 & 31 \\ 1 & 103 \end{vmatrix}}{1} = -72$    | $\frac{-\begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix}}{1} = 0$     | $\frac{-\begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix}}{1} = 0$     |
| $s^0$ | $\frac{-\begin{vmatrix} 1 & 103 \\ -72 & 0 \end{vmatrix}}{-72} = 103$ | $\frac{-\begin{vmatrix} 1 & 0 \\ -72 & 0 \end{vmatrix}}{-72} = 0$ | $\frac{-\begin{vmatrix} 1 & 0 \\ -72 & 0 \end{vmatrix}}{-72} = 0$ |

Total number of poles=3

Number of sign changes in first column= 2

Number of poles in RHP=2

Number of poles in LHP=1



$$\frac{C(s)}{R(s)} = \frac{200}{s^4 + 6s^3 + 11s^2 + 6s + 200}$$

$$s^4 \quad \quad \quad 1 \quad \quad \quad 11 \quad \quad \quad 200$$

$$s^3 \quad \quad \quad 6 \quad \quad \quad 6$$

$$s^2 \quad - \quad \frac{\begin{vmatrix} 1 & 11 \\ 6 & 6 \end{vmatrix}}{6} = 10 \quad 200$$

-(6-66)

$$s \quad - \quad \frac{\begin{vmatrix} 6 & 6 \\ 10 & 200 \end{vmatrix}}{10} = -114 \quad \frac{\begin{vmatrix} 6 & 0 \\ 10 & 0 \end{vmatrix}}{10} = 0$$

$$s^0 \quad 200$$

Example

→ 2 sign changes so two roots in RHP.

→ system is unstable.

# Example

**TABLE 6.10** Routh table for Example 6.6

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|       |               |     |                |    |     |
|-------|---------------|-----|----------------|----|-----|
| $s^4$ |               | 1   |                | 11 | 200 |
| $s^3$ | <del>6</del>  | 1   | <del>6</del>   | 1  |     |
| $s^2$ | <del>10</del> | 1   | <del>200</del> | 20 |     |
| $s^1$ |               | -19 |                |    |     |
| $s^0$ |               | 20  |                |    |     |

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## Routh Hurwitz Criterion: Special Cases

1. Zero only in the first column of a row
2. Entire row is zero

# Routh Hurwitz Criterion: Special Case

- Zero only in the first column of a row:

If the first element of a row is zero, division by zero would be required to form the next row. To avoid this phenomenon, an epsilon,  $\epsilon$ , is assigned to replace the zero in the first column. The value  $\epsilon$  is then allowed to approach zero from either the positive or the negative side, after which the signs of the entries in the first column can be determined.

**Example 6.2: Determine the stability of the closed-loop transfer function.**

$$T(s) = \frac{10}{s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 3}$$

Completed Routh table

|       |                                                         |               |   |
|-------|---------------------------------------------------------|---------------|---|
| $s^5$ | 1                                                       | 3             | 5 |
| $s^4$ | 2                                                       | 6             | 3 |
| $s^3$ | $\cancel{0} \epsilon$                                   | $\frac{7}{2}$ | 0 |
| $s^2$ | $\frac{6\epsilon - 7}{\epsilon}$                        | 3             | 0 |
| $s^1$ | $\frac{42\epsilon - 49 - 6\epsilon^2}{12\epsilon - 14}$ | 0             | 0 |
| $s^0$ | 3                                                       | 0             | 0 |

**Example 6.2: Determine the stability of the closed-loop transfer function.**

| Label | First Column                                            | $\epsilon = +$ | $\epsilon = -$ |
|-------|---------------------------------------------------------|----------------|----------------|
| $s^5$ | 1                                                       | +              | +              |
| $s^4$ | 2                                                       | +              | +              |
| $s^3$ | $\cancel{0} \ \epsilon$                                 | +              | -              |
| $s^2$ | $\frac{6\epsilon - 7}{\epsilon}$                        | -              | +              |
| $s^1$ | $\frac{42\epsilon - 49 - 6\epsilon^2}{12\epsilon - 14}$ | +              | +              |
| $s^0$ | 3                                                       | +              | +              |

**Table 6.5**

Determining signs in first column of a Routh table with zero as first element in a row

# Example

| Example $T(s) = \frac{1}{2s^5 + 3s^4 + 2s^3 + 3s^2 + 2s + 1}$ |                                                        |                                  |   |                              |
|---------------------------------------------------------------|--------------------------------------------------------|----------------------------------|---|------------------------------|
| $s^5$                                                         | 2                                                      | 2                                | 2 | $\rightarrow 2$ sign changes |
| $s^4$                                                         | 3                                                      | 3                                | 1 | $\rightarrow 2$ Roots in RHP |
| $s^3$                                                         | $0 \text{ } \epsilon^{+ve}$                            | $+(2-\frac{6}{3}) = \frac{4}{3}$ |   | $\rightarrow 3$ Roots in LHP |
| $s^2$                                                         | $-(\frac{4-3\epsilon}{\epsilon})$                      | 1                                |   |                              |
| $s$                                                           | $\frac{12\epsilon - 16 - 3\epsilon^2}{9\epsilon - 12}$ |                                  |   |                              |
| $s^0$                                                         | 1                                                      |                                  |   |                              |